

Shortness of Breath

(Dyspnea)

By **Rebecca Dezube**, MD, MHS, Johns Hopkins University
Reviewed/Revised Nov 2025

Shortness of breath—what doctors call dyspnea—is the unpleasant sensation of having difficulty breathing. People experience and describe shortness of breath differently depending on the cause.

The rate and depth of breathing normally increase during exercise and at high altitudes, but the increase seldom causes discomfort. Breathing rate is also increased at rest in people with many disorders, whether of the lungs or other parts of the body. For example, people with a fever generally breathe faster.

With dyspnea, faster breathing is accompanied by the sensation of running out of air. People feel as if they cannot breathe fast enough or deeply enough. They may notice that more effort is needed to expand the chest when breathing in or to expel air when breathing out. They may also have the uncomfortable sensation that inhaling (inspiration) is urgently needed before exhaling (expiration) is completed and have various sensations often described as tightness in the chest.

Other symptoms, such as [cough](#) or [chest pain](#), may be present depending on the cause of dyspnea.

Causes of Shortness of Breath

Dyspnea is usually caused by disorders of the lungs or heart (see table

[Some Causes and Features of Shortness of Breath](#))

The **most common** causes overall include:

- [Asthma](#)
- [Pneumonia](#)

- [Chronic obstructive pulmonary disease](#) (COPD)
- A [heart attack](#) or [angina](#) (chest pain due to inadequate blood flow and oxygen to the heart—called myocardial ischemia)
- [Heart failure](#)
- Pulmonary hypertension
- Physical deconditioning (for example, weakening of muscles and the heart due to inactivity)
- Weight gain
- Pregnancy

[Pulmonary embolism](#) (sudden blockage of an artery of the lung, usually by a blood clot) is a less common, but serious cause.

The most common cause in people with a chronic lung or heart disorder is worsening of their disease.

However, such people may also develop another disorder. For example, people with long-standing asthma may have a heart attack, or people with chronic heart failure may develop pneumonia.

Lung disorders

People who have lung disorders often experience dyspnea when they physically exert themselves. During exercise, the body makes more carbon dioxide and uses more oxygen. The respiratory center in the brain speeds up breathing when blood levels of oxygen are low or blood levels of carbon dioxide are high. If the heart or lungs are not functioning normally, even a little exertion can dramatically increase the breathing rate and dyspnea. Dyspnea can be so unpleasant that people avoid exertion. As the lung disorder becomes more severe, dyspnea may occur even when the person is at rest.

Dyspnea may result from:

- Restrictive lung disorders
- Obstructive lung disorders

In **restrictive lung disorders** (such as [idiopathic pulmonary fibrosis](#)), lungs become stiff and require more effort to expand during inhalation. Severe curvature of the spine ([scoliosis](#)) can also restrict breathing because it reduces movement of the rib cage.

In **obstructive disorders** (such as [COPD](#) or [asthma](#)), resistance to airflow is increased because the airways are narrowed. Because airways widen during inhalation, air can

usually be pulled in. However, because airways narrow during exhalation, air cannot be exhaled from the lungs as fast as normal, and people wheeze and breathing is labored. Dyspnea results when too much air is left in the lungs after exhaling.

People with asthma have dyspnea when they have an attack. Doctors typically advise people to keep an inhaler on hand to use during an attack. The medication in the inhaler helps open the airways.

Heart failure

The heart pumps blood through the lungs. If the heart is pumping inadequately (called [heart failure](#)), fluid may accumulate in the lungs—a disorder called pulmonary edema. This disorder causes dyspnea that is often accompanied by a feeling of smothering or heaviness in the chest. The fluid accumulation in the lungs may also narrow the airways and cause wheezing—a disorder called [cardiac asthma](#).

Some people with heart failure have orthopnea, paroxysmal nocturnal dyspnea, or both. Orthopnea is shortness of breath that occurs when people lie down and is relieved by sitting up. Paroxysmal nocturnal dyspnea is a sudden, often terrifying attack of dyspnea during sleep. People awaken gasping and must sit or stand to catch their breath. This disorder is an extreme form of orthopnea and a sign of severe heart failure.

Anemia

When people have [anemia](#) or have lost a large amount of blood because of an injury, they have fewer red blood cells. Red blood cells carry oxygen to the tissues, so in these people, the amount of oxygen that blood can deliver is decreased. Most people with anemia are comfortable sitting still. However, they often feel dyspnea during physical activity because the blood cannot deliver the increased oxygen the body requires. Thus, they breathe rapidly and deeply in a reflex effort to try to increase the amount of oxygen in the blood.

Other causes

If a large amount of acid accumulates in the blood (called [metabolic acidosis](#)), people may feel out of breath and begin to pant quickly. Severe kidney failure, sudden worsening of diabetes mellitus, and ingestion of certain medications or poisons (such as methanol and ethylene glycol) can cause metabolic acidosis. Anemia and heart failure may contribute to dyspnea in people with kidney failure.

In **hyperventilation syndrome**, people feel that they cannot get enough air, and they breathe heavily and rapidly. This syndrome is commonly caused by anxiety rather

than a physical problem. Many people who experience it are frightened, may have chest pain, and may believe they are having a heart attack. They may have a change in consciousness, usually described as feeling that events occurring around them are far away, and they may feel tingling in their hands and feet and around their mouth.

Evaluation of Shortness of Breath

The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

In people with dyspnea, the following symptoms are of particular concern:

- Shortness of breath at rest
- Agitation or confusion or a decreased level of consciousness
- Chest discomfort or the feeling the heart is pounding or racing or has skipped a beat (palpitations)
- Weight loss
- Night sweats

When to see a doctor

People who have shortness of breath at rest, chest pain, palpitations, agitation, confusion or a decreased level of consciousness or have difficulty moving air in or out of their lungs should go to the hospital right away. Such people may need immediate testing, treatment, and often admission to the hospital. Other people should call a doctor. The doctor can determine how rapidly they need to be evaluated based on the nature and severity of their symptoms, their age, and any underlying medical conditions. Typically, they should be evaluated as soon as possible.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What doctors find during the history and physical examination often suggests a cause and the tests that may need to be done (see table

Some Causes and Features of Shortness of Breath

Doctors ask questions to determine:

- When shortness of breath started

- Whether it started abruptly or gradually
- How long the person has felt short of breath
- Whether any conditions (such as cold, exertion, exposure to allergens, or lying down) trigger it or make it worse

The person is also asked questions about past medical history (including any lung or heart disorders), a history of smoking, any family members who have had high blood pressure or high cholesterol levels, and risk factors for pulmonary embolism (such as recent hospitalization, surgery, or long-distance travel).

The physical examination focuses on the heart and lungs. Doctors listen to the lungs for congestion, wheezing, and abnormal sounds called crackles. They listen to the heart for murmurs (suggesting a heart valve disorder). Swelling of both legs suggests heart failure, but swelling of only one leg is more likely to result from a blood clot in the leg. A blood clot in the leg may break off and travel to the blood vessels in the lungs, causing pulmonary embolism.



Some Causes and Features of Shortness of Breath

Cause	Common Features*	Tests†
Acute (develops within minutes or hours)		
<u>Anxiety disorder</u> — hyperventilation	Shortness of breath related to a specific situation, often accompanied by agitation and tingling or numbness in the fingers and/or around the mouth Normal results on the heart and lung examination	A doctor's examination Sometimes one or more of the following tests: • Tests to evaluate how well the lungs are functioning (<u>pulmonary function testing</u>), or measurement of peak air flow (how fast air can be exhaled)
<u>Asthma</u> ‡	Wheezing that starts spontaneously or after exposure to specific stimuli (such as pollen or another allergen, an upper respiratory infection, cold air, or exercise) Usually a history of asthma	• Measurement of lung function before and after exercise or administration of methacholine (a drug that narrows airways)

